Chapter 1

Relying on AI to Improve Human Interaction

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People interact with each other in myriad ways. Few people realize, in fact, just how many different ways communication occurs. When many people think about communication, they think about writing or talking. However, interaction can take many other forms, including eye contact, tonal quality, and even scent.

An example of the computer version of enhanced human interaction is the electronic "nose," which relies on a combination of electronics, biochemistry, and artificial intelligence to perform its task and has been applied to a wide range of industrial applications and research (see `tinyurl.com/488jfzut`). In fact, the electronic nose can even "sniff out" diseases (see `tinyurl.com/28cfcjek`).

This chapter concentrates more along the lines of standard communication, however, including body language. You gain a better understanding

of how AI can enhance human communication through means that are less costly than building your own electronic nose.

AI can also enhance the manner in which people exchange ideas. In some cases, AI provides entirely new methods of communication, but in many cases, AI provides a subtle (or sometimes not so subtle) method of enhancing existing ways to exchange ideas. Humans rely on exchanging ideas to create new technologies, build on existing technologies, or learn about technologies needed to increase an individual's knowledge. Ideas are abstract, which makes exchanging them particularly difficult at times, so AI can provide a needed bridge between people.

At one time, if someone wanted to store their knowledge to share with someone else, they generally relied on writing. In some cases, they could also augment their communication by using graphics of various types. However, only some people can use these two forms of media to gain new knowledge; many people require more, which is why online sources such as YouTube have become hugely popular. Interestingly enough, you can augment the power of multimedia, which is already substantial, by using AI, and this chapter tells you how.

This chapter also helps you understand how an AI can give you almost superhuman sensory perception. Perhaps you want that electronic nose after all; it does provide significant advantages in detecting scents that are significantly less aromatic than humans can smell. Imagine being able to smell at the same level as a dog does (which uses 100 million aroma receptors, versus the 1 million aroma receptors that humans possess). It turns out there are two ways to achieve this goal: monitors that a human accesses indirectly and direct stimulation of human sensory perception.

Developing New Ways to Communicate

Communication involving a developed language initially took place between humans via the spoken versus written word. The only problem with spoken communication is that the two parties must appear near enough together to talk. Consequently, written communication is superior in many respects because it allows time-delayed communication that

doesn't require the two parties to ever see each other. The three main methods of human nonverbal communication rely on

- **Alphabets/Iconographs:** The abstraction of components of human words or symbols
- **Language:** The stringing together of words or symbols to create sentences or convey ideas in written form
- **Body language:** The augmentation of language with context

The first two methods are direct abstractions of the spoken word. They aren't always easy to implement, but people have been implementing them for thousands of years. The body-language component is the hardest to implement because you're trying to create an abstraction of a physical process. Writing helps convey body language using specific terminology, such as that described at `tinyurl.com/27newjbf`. However, the written word falls short, so people augment it with symbols, such as emoticons and emojis (you can read about their differences at `tinyurl.com/9jyc5n4n`). The following sections discuss these issues in more detail.

Creating new alphabets

The introduction to this section mentions two new alphabets used in the computer age: emoticons and emojis (`tinyurl.com/wjsw8te5` and `emojipedia.org`). The sites where you find these two graphical alphabets online can list hundreds of them. For the most part, humans can interpret these iconic alphabets without much trouble because the alphabets resemble facial expressions; an application lacks the human sense of art, however, so computers often require an AI just to figure out which emotion a human is trying to convey by using the little pictures. Fortunately, you can find standardized lists, such as the Unicode emoji chart at `tinyurl.com/j4bdmm3m`.

Emoticons are an older technology, and many people are trying their best to forget them (but they likely won't succeed because emoticons are easy to type, though you might recall the movie titled *The Emoji Movie*). You can also turn your selfies into emojis (see "The 7 Best Free Apps to Turn Selfies Into Emojis, Stickers, and More" at `MakeUseOf.com`). Many peo-

ple have a hard time figuring out emojis, so you can check Emojipedia to see what they mean.



REMEMBER Humans have created new alphabets to meet specific needs since the beginning of the written word. Emoticons and emojis represent two of many alphabets that you can count on humans creating as the result of the internet and the use of AI. In fact, it may actually require an AI to keep up with them all. However, it's equally important to remember that some characters are lost as time progresses. For example, check out the article "10 Letters That Didn't Make the Alphabet" at `https://tinyurl.com/zzsjzcfc`.

Working with emojis and other meaningful graphics

Many text references are now sprinkled with emojis and other iconography. Most of the references you see online today deal with emojis and emoticons, either removing them or converting them to text.

It's not uncommon to find bits and pieces of other languages sprinkled throughout a text, and these words or phrases need to be handled in a meaningful way. The problem with translating certain languages into a form in which they can act as input to a natural language processing (NLP) AI model is that the concept of the language differs from English. For example, when working with Chinese, you deal with ideas rather than with pronunciation, as you do with English.

Some situations also require that you process meaningful graphics because part of the text meaning is in the graphic. This sort of translation need commonly arises in technical or medical texts. The article "How Image Analysis and Natural Language Processing Can Be Combined to Improve Precision Medicine," by Obi Igbokwe, at `https://tinyurl.com/k7djha34`, discusses how to accomplish this task.



REMEMBER The point of these various translations of nontext into a textual form is that humans communicate in many ways, and AI can help make such communication easier and improve comprehension. In addition, using AI to perform NLP makes it possible to look for patterns, even in text that is heavily imbued with nontext elements.

Automating language translation

The world has always had a problem with the lack of a common language. Yes, English has become pervasive — though it's still not universal. Having someone translate between languages can be expensive, cumbersome, and error-prone, so translators, though necessary in many situations, aren't necessarily the ideal answer, either. For those who lack the assistance of a translator, dealing with other languages can be quite difficult, which is where applications such as Google Translate come into play.

Google Translate offers to automatically detect the language for you. What's interesting about this feature is that it works extremely well in most cases. Part of the responsibility for this feature is the Google Neural Machine Translation (GNMT) system. It can examine entire sentences to make sense of them and provide better translations than applications that use phrases or words as the basis for creating a translation (see `tinyurl.com/8by975xx` for details).



TECHNICAL STUFF What is even more impressive is that GNMT can translate between languages even when it has no specific translator, using an artificial language, an *interlingua.* However, you should realize that an interlingua doesn't function as a universal translator; it's more of a universal bridge. Say that the GNMT doesn't know how to translate between Chinese and Spanish. However, it can translate between Chinese and English and between English and Spanish. By building a 3D network representing these three languages (the interlingua), GNMT is able to create its own translation between Chinese and Spanish. Unfortunately, this sys-

tem doesn't work for translating between Chinese and whale language because no method is available yet to understand and translate whale language in any human language. Humans still need to create a base translation for GNMT to do its work.

Incorporating body language

A significant part of human communication occurs with body language, which is why the use of emoticons and emojis is important. However, people are becoming more used to working directly with cameras to create videos and other forms of communication that involve no writing. In this case, a computer could possibly listen to human input, parse it into tokens representing human speech, and then process those tokens to fulfill a request, similar to the way Alexa and Google Home (and others) work.



REMEMBER Unfortunately, merely translating the spoken word into tokens won't do the job, because the whole issue of nonverbal communication remains. In this case, the AI must be able to read body language directly. The article "Computer Reads Body Language" (`https://tinyurl.com/49yrz3s5`) from Carnegie Mellon University discusses some of the issues that developers must solve to make reading body language possible. The picture at the beginning of the article gives you some idea of how the computer camera must capture human positions to read the body language, and the AI often requires input from multiple cameras to make up for such issues as having part of the human anatomy obscured from the view of a single camera. The reading of body language involves interpreting these human characteristics:

- Posture
- Head motion
- Facial expression
- Eye contact
- Gestures

Of course, other characteristics must be considered, but if an AI can even get these five areas down, it can go a long way toward providing a correct body-language interpretation. In addition to body language, current AI implementations consider characteristics like tonal quality, which makes for an extremely complex AI that still doesn't come close to doing what the human brain does seemingly without effort.



After an AI can read body language, it must also provide a means to output it when interacting with humans. Given that reading body language (facial expressions, body position, placement of hands, and other factors) is in its infancy, robotic or graphical presentation of body language is even less developed.

Exchanging Ideas

An AI doesn't have ideas because it lacks both intrapersonal intelligence and the ability to understand. However, an AI can enable humans to exchange ideas in a manner that creates a whole that is greater than the sum of its parts. In many cases, the AI isn't performing any sort of exchange. Instead, the humans involved in the process perform the exchange by relying on the AI to augment the communication process. The following sections provide additional details about how this process occurs.

Creating connections

A human can exchange ideas with another human, but only as long as the two humans know about each other. The problem is that many experts in a particular field don't actually know each other — at least not well enough to communicate effectively. An AI can perform research based on the flow of ideas that a human provides and then create connections with other humans who have that same (or similar) flow of ideas.

One way in which this type of convergence occurs is in social media sites such as LinkedIn, where the idea is to create connections between people

based on a number of criteria. A person's network becomes the means by which LinkedIn uses AI to suggest other potential connections. Ultimately, the purpose of these connections from the user's perspective is to gain access to new human resources, make business contacts, create a sale, or perform other tasks that LinkedIn enables.

Augmenting communication

To exchange ideas successfully, two humans need to communicate well. The only problem is that humans sometimes don't communicate well, and sometimes they don't communicate at all. The issue is one of translating not only words but also ideas. The societal and personal biases of individuals can preclude the communication because an idea for one group may not translate at all for another group. For example, the laws in one country might make someone think in one way, but the laws in another country could make another person think in an entirely different manner.

Theoretically, an AI could help communication between disparate groups in numerous ways. Of course, language translation (assuming that the translation is accurate) is one of these methods. However, an AI could provide cues to what is and isn't culturally acceptable by prescreening materials. Using categorization, an AI could also suggest aids like alternative graphics and other elements to help communication take place in a manner that helps both parties. For example, you could replace an image of a Red Cross with a Red Crescent or both to represent first aid in different cultures.

Defining trends

Humans often base ideas on trends. To visualize how the idea works, however, other parties in the exchange of ideas must also see those trends, and communicating using this sort of information is notoriously difficult. AI can perform various levels of data analysis and present the output graphically. The AI can analyze the data in more ways and more quickly than a human can, so the story the data tells is specifically the one

you need it to tell. The data remain the same; the presentation and interpretation of the data change.

Studies show that humans relate better to graphical output than to tabular output, and graphical output definitely makes trends easier to see. You generally use tabular data to present only specific information; graphics always work best for showing trends. Using AI-driven applications can also make creating the right sort of graphical output for a particular requirement easier. Not all humans see graphics in precisely the same way, so matching a graphical type to your audience is essential.

Using Multimedia

Most people learn by using multiple senses and multiple approaches. A doorway to learning that works for one person may leave another completely mystified. Consequently, the more ways in which a person can communicate concepts and ideas, the more likely it is that other people will understand what the person is trying to communicate. Multimedia normally consists of sound, graphics, text, and animation, though some multimedia does more.

AI can help with multimedia in numerous ways. One of the most important involves the creation, or *authoring*, of the multimedia. You find AI in applications that help with everything from media development to media presentation. For example, when translating the colors in an image, an AI may provide the benefit of helping you visualize the effects of those changes faster than trying one color combination at a time (the brute force approach).

MULTIMEDIA AND ADDRESSING PEOPLE'S FUNCTIONAL NEEDS

Most people have some particular functional need relating to how they take in and understand information. Considering such needs as part of people's use of multimedia is important. The whole intent of multimedia is to communicate ideas in as many ways as possible so that just about everyone can understand the ideas and concepts you want to present.

Even when a presentation as a whole uses multimedia successfully, individual ideas can become lost when the presentation uses only a single method to communicate them. For example, communicating a sound only aurally (instead of adding closed captioning to a visual presentation, for example) almost guarantees that only those with excellent hearing will receive the idea. A subset of those with the required hearing level still won't get the idea because it may appear as only so much noise to them, or they simply won't learn through the limited method offered in the presentation.

Using as many methods as possible to communicate each idea is essential if you want to reach as many people as possible. Even if you get the information in one way, also getting it another way provides useful confirmation that you understood correctly.

After using multimedia to present ideas in more than one form, those receiving the ideas must process the information. A secondary use of AI relies on the use of neural networks to process the information in various ways. Categorizing the multimedia is now an essential use of the technology. However, in the future, you can look forward to using AI to help in 3D reconstruction of scenes based on 2D pictures. Imagine police personnel being able to walk through a virtual crime scene with every detail faithfully captured.

People used to speculate that various kinds of multimedia would appear in new forms. For example, imagine a newspaper that provides Harry Potter-like dynamic displays. Most of the technological pieces are available today, but the issue comes down to the market: For a technology to become successful, it must have a market — that is, a means for paying for itself.

Embellishing Human Sensory Perception

One way that AI truly excels at improving human interaction is by augmenting humans in one of two ways: by allowing them to use their native senses to work with augmented data or by augmenting the native senses

to do more. The following sections discuss both approaches to enhancing human sensing and, therefore, improving communication.

Shifting data spectrum

When performing various kinds of information gathering, humans often employ technologies that filter or shift the data spectrum with regard to color, sound, touch, or smell. Humans still use native capabilities, but some technology changes the input such that it works with that native capability. One of the most common examples of spectrum shifting is astronomy, in which shifting and filtering light enables people to see astronomical elements, such as nebula, in ways that the naked eye can't — thereby improving our understanding of the universe.

Teaching a robot to feel by touch is in its infancy, as described at `tinyurl.com/ukjscsye` and `tinyurl.com/y998s2h8`. Most efforts now focus on helping the robot work better by using tactile responses as part of manipulating its machinery, such as the light touch needed to grasp an egg versus the heavier touch required to lift a barbell. As this technology moves far enough forward, it might become possible for various AIs to communicate with humans via direct touch or the description of various kinds of touch.

Shifting and filtering colors, sounds, touches, and smells manually can require a great deal of time, and the results can disappoint even when performed expertly, which is where AI comes into play. An AI can try various combinations far faster than a human can and can locate the potentially useful combinations with greater ease because an AI performs the task in a consistent manner.



The most intriguing technique for exploring our world, however, is completely different from what most people expect. What if you could smell a color or see a sound? The occurrence of *synesthesia*, which is the use of one sense to interpret input from a different sense, is well documented in humans — see `tinyurl.com/2s9tb284`. For more about the

general understanding of the current state of synesthesia, see `www.psychologytoday.com/us/basics/synesthesia`.

Augmenting human senses

As an alternative to using an external application to shift a data spectrum and somehow make that shifted data available for use by humans, you can augment human senses. In augmentation, a device, either external or implanted, enables a human to directly process sensory input in a new way. The idea is an old one: Use tools to make humans ever more effective at performing an array of tasks. In this scenario, humans receive these two forms of augmentation:

- ***Physical augmentation*** of human senses already takes place in many ways, and it's guaranteed to increase as humans become more receptive to various kinds of implants. For example, night vision glasses now allow humans to see at low light levels, with high-end models providing color vision controlled by a specially designed processor.
  In the future, eye augmentation or replacement may allow people to see any part of the spectrum. This augmentation or replacement would be controlled by the person's thoughts so that people would see only that part of the spectrum needed to perform a specific task.
- ***Intelligence augmentation*** requires more intrusive measures but also promises to allow humans to exercise far greater capabilities. Unlike AI, intelligence augmentation (IA) has a human actor at the center of the processing — the human provides the creativity and intent that AI now lacks.